INTRODUCTION

1.INTRODUCTION

Neostigmine Methyl sulphate Injection, USP is the dimethylcarbamate of (m-hydroxyphenyl) trimethylammonium methylsulfate. Neostigmine Methylsulfate (neostigmine methylsulfate injection), an anticholinesterase agent, is a bitter tasting, white crystalline powder and is very soluble in water and soluble in alcohol. Neostigmine Methylsulfate (neostigmine methylsulfate (neostigmine methylsulfate injection) injection) Injection, USP is a sterile, nonpyrogenic solution intended for intramuscular, subcutaneous or slow intravenous use.

1.1 INDICATION

Neostigmine Methylsulfate Injection, USP is indicated for:

- The symptomatic control of myasthenia gravis when oral therapy is impractical.
- The prevention and treatment of postoperative distention and urinary retention after mechanical obstruction has been excluded.
- Reversal of effects of non-depolarizing neuromuscular blocking agents (e.g., tubocurarine, metocurine, gallamine or pancuronium) after surgery.
- All of these problems contribute to the development of pimples. A zit appears when bacteria grow in a clogged pore and the oil is unable to escape.

1.2. INTRODUCTION TO ANALYTICAL METHOD

Pharmaceutical products formulated with more than one drug, typically referred to as combination products. These combination products can present daunting challenges to the analytical chemist responsible for the development and validation of analytical methods. The development and validation of analytical methods [Spectrophotometric, High performance liquid chromatography (HPLC) & High performance thin layer chromatography (HPTLC)] for drug products containing more than one active ingredient. The official test methods that result from these processes are used by quality control laboratories to ensure the identity, purity, potency. and performance of drug products.

The number of drugs introduced into the market is increasing every year. These drugs may be either new entities or partial structural modification of the existing ones. Very often there is a time lag from the date of introduction of a drug into the market to the date of its inclusion in pharmacopoeias. This happens because of the possible uncertainties in the continuous and wider usage of these drugs, reports of new toxicities (resulting in their withdrawal from the market), development of patient resistance and introduction of better drugs by competitors. Under these conditions, standards and analytical procedures for these drugs may not be available in the pharmacopoeias. It becomes necessary, therefore to develop newer analytical methods for such drugs.

1.3 INTRODUCTION TO HPLC METHOD

Liquid chromatography (LC) is a physical separation technique conducted in the liquid phase. A sample is separated into its constituent components (or analytes) by distributing between the mobile phase (a flowing liquid) and a stationary phase (sorbents packed inside a column). For example, the flowing liquid can be an organic solvent forced through the column at high speed and the stationary phase can be porous silica particles packed in a column. The modern form of column chromatography has been called high performance, high Pressure, high- resolution and high-speed liquid chromatography. HPLC is a modern form of LC that uses small- particle columns through which the mobile phase is pumped at high pressure.

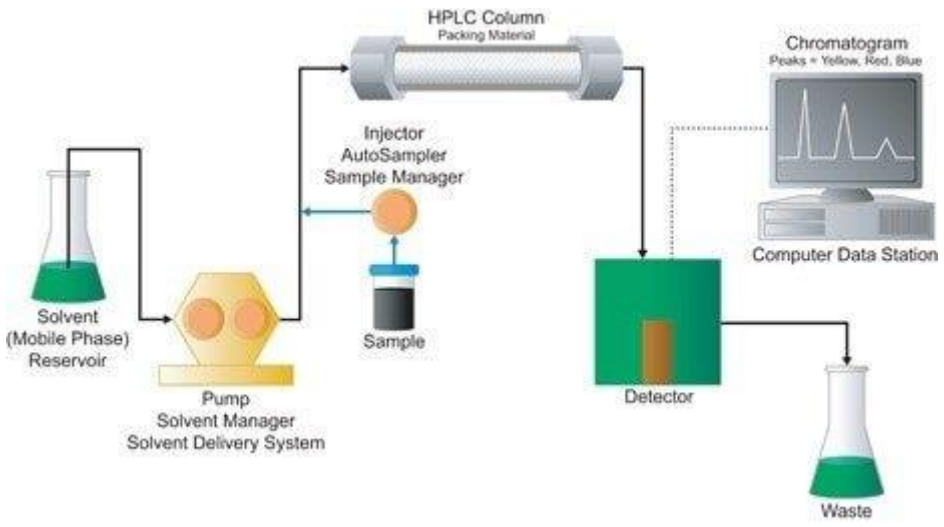


Fig.1.1 Block Diagram of HPLC

High-performance liquid chromatography (HPLC), sometimes called high-pressure liquid chromatography, is a separation technique based on a solid stationary phase and a liquid mobile phase. It describes the work out flow about High performance liquid chromatography (HPLC).

It relies on high pressure pumps, which deliver mixtures of various solvents, called the mobile phase, which flows through the system, collecting the sample mixture on the way, delivering it into a cylinder, called the column, filled with solid particles, made of adsorbent material, called the stationary phase.

Each component in the sample interacts differently with the adsorbent material, causing different migration rates for each component. These different rates lead to separation as the species flow out of the column into a specific detector such as UV detectors. The output of the detector is a graph, called a chromatogram. Chromatograms are graphical representations of the signal intensity versus time or volume, showing peaks, which represent components of the sample. Each sample appears in its respective time, called its retention time, having area proportional to its amount.

HPLC is widely used for manufacturing (*e.g.*, during the production process of pharmaceutical and biological products), legal (*e.g.*, detecting performance enhancement drugs in urine), research (*e.g.*, separating the components of a complex biological sample, or of similar synthetic chemicals from each other), and medical (*e.g.*, detecting vitamin D levels in blood serum) purposes.

Chromatography can be described as a mass transfer process involving adsorption and/or partition. As mentioned, HPLC relies on pumps to pass a pressurized liquid and a sample mixture through a column filled with adsorbent, leading to the separation of the sample components. The active component of the column, the adsorbent, is typically a granular material made of solid particles (*e.g.*, silica, polymers, etc.), 1.5–50 μm in size, on which various reagents can be bonded.

The components of the sample mixture are separated from each other due to their different degrees of interaction with the adsorbent particles. The pressurized liquid is typically a mixture of solvents (*e.g.*, water, buffers, acetonitrile and/or methanol) and is referred to as a

"mobile phase". Its composition and temperature play a major role in the separation process by influencing the interactions taking place between sample components and adsorbent. These interactions are physical in nature, such as hydrophobic (dispersive), dipole–dipole and ionic, most often a combination.

1.3.1. Principle of separation

The principle of separation in normal phase mode and reverse phase mode is adsorption. When mixtures of components are introduced in to a HPLC column, they travel according to their relative affinities towards the stationary phase. The component which has more affinity towards the adsorbent travels slower. The component which has less affinity towards the stationary phase travels faster. Since no two components have the same affinity towards the stationary phase, the components are separated. The specific intermolecular interactions between the molecules of a sample and the packing material define their time “on-column”. Hence, different constituents of a sample are eluted at different times. Thereby, the separation of the sample ingredients is achieved.

There are different modes of separation in HPLC:

1) Normal phase mode.

2) Reversed phase mode.

3) Ion exchange chromatography.

4) Reverse phase ion pair chromatography.

5) Affinity chromatography and

6) Size exclusion chromatography.

1.Normal phase:

In the normal phase mode, the stationary phase is polar and the mobile phase is non polar in nature. In this technique, non-polar compounds travel faster and are eluted first. This is because of the lower affinity between the non-polar compounds and the stationary phase. Polar compounds are retained for longer times because of their higher affinity with the stationary phase. These compounds, therefore take more times to elute. Normal phase mode of separation is therefore, not generally used for pharmaceutical applications because most of the drug molecules are polar in nature and hence take longer time to elute.

Although this type of chromatography isn't used as often, there are many good reasons to choose Normal Phase HPLC. It is great when a compound is too hydrophobic or hydrophilic for reverse-phase HPLC. It can also be used for isomer separation, if the sample injection solvent is non-polar, or if recovery in non-polar solvents is desirable.

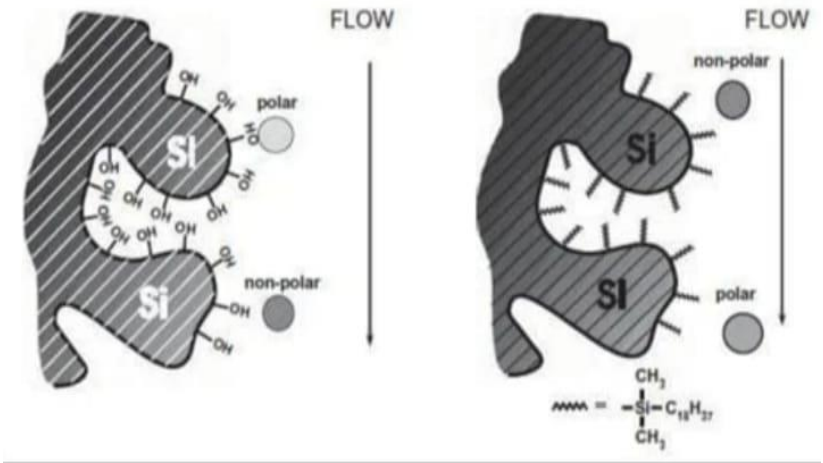


Fig.1.2 Schematic diagram depicting separation modes of

(a) Normal-phase chromatography (NPC) and
(b) Reversed-phase chromatography (RPC)

2) Reversed phase mode:

Reversed phase mode is the most popular mode for analytical and preparative separations of compound of interest in chemical, biological, pharmaceutical, food and biomedical sciences. In this mode, the stationary phase is non polar hydrophobic packing with octyl or octa decyl functional group bonded to silica gel and the mobile phase is polar solvent. An aqueous mobile phase allows the use of secondary solute chemical equilibrium (such as ionization control, ion suppression, ion pairing and complexation) to control retention and selectivity. The polar compound gets eluted first in this mode and non-polar compounds are retained for longer time. As most of the drugs and pharmaceuticals are polar in nature, they are not retained for longer times and hence elute faster.

The different columns used are octa decyl silane (ODS) or Cis, Cx, C4, etc., (in the order of increasing polarity of the stationary phase). In order to retain the organic components in mixtures, the stationary phases, packed within columns, consist of a hydrophobic substrates, bonded to the surface of porous silica-gel particles in various geometries (spheric, irregular), at different diameters (sub-2, 3, 5, 7, 10 um), with varying pore diameters (60, 100, 150, 300). The particle's surface is covered by chemically bonded hydrocarbons, such as C3, C4, C8, C18 and more. The longer the hydrocarbon associated with the stationary phase, the longer the sample components will be retained.

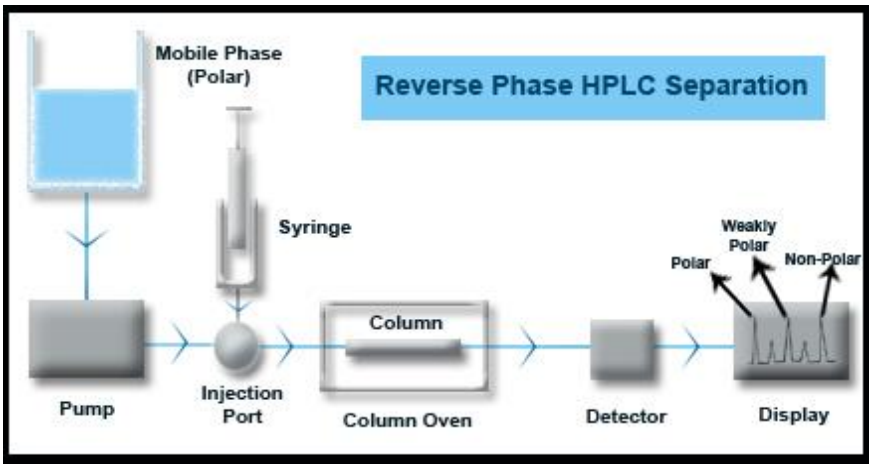


Fig.1.3 Reversed Phase Chromatography

3) Ion exchange chromatography:

Ion exchange chromatography (or ion chromatography) is a process that allows the separation of ions and polar molecules based on their affinity to ion exchangers. The principle of separation is thus by reversible exchange of ions between the target ions present in the sample solution to the ions present on ion exchangers.

In ion exchange chromatography, the stationary phase contains ionic groups like NR3 or SO, which interact with the ionic groups of the sample molecules. This is suitable for the separation of charged molecules only. Changing the pH and salt concentration can modulate the retention.

In this process, two types of exchangers i.e., cationic and anionic exchangers can be used.

1. **Cationic exchangers** possess negatively charged group, and these will attract positively charged cations. These exchangers are also called “Acidic ion exchange” materials, because their negative charges result from the ionization of acidic group.
2. **Anionic exchangers** have positively charged groups that will attract negatively charged anions. These are also called “Basic ion exchange” materials.

- Ion exchange chromatography is most often performed in the form of column chromatography. However, there are also thin-layer chromatographic methods that work basically based on the principle of ion exchange.

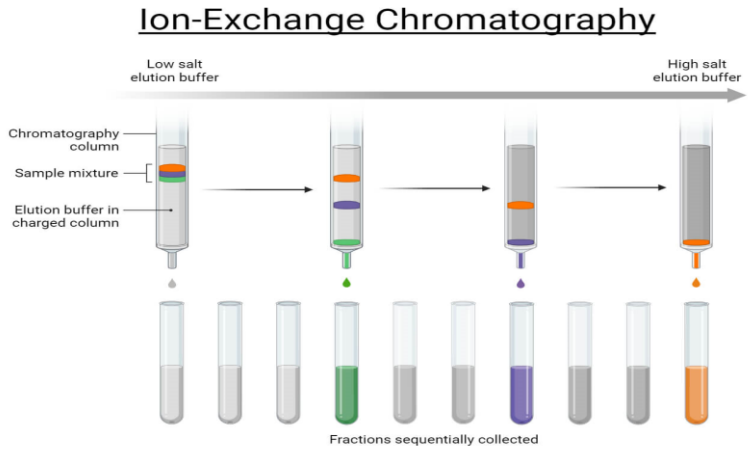


Fig.1.4 Ion exchange chromatography

4) Reverse phase ion pair chromatography:

Ion pair chromatography may be used for the separation of ionic compounds and this method can also substitute for ion exchange chromatography. Strong acidic and basic compounds may be separated by reversed phase mode by forming ion pairs (coulumbic association species formed between two ions of opposite electric charge) with suitable counter ions. This technique is referred to as reversed phase ion pair chromatography or soap chromatography.

Ion pair chromatography helps separate charged analytes using the popular reverse phase chromatography. Ion pair reagents enhance peak shape and retention time when modifying mobile phase ratios or changing stationary phase is not of much help.

Ion exchange chromatography had been in existence for several years but introduction of Ion pair chromatography offered several advantages:

- Simple preparation of buffers.
- Wide choice of carbon chain lengths for improved retention and separation.
- Significantly reduced separation time.
- Simultaneous separation of both ionized and non-ionized molecules.
- High reproducibility of results.
- Improved peak shapes

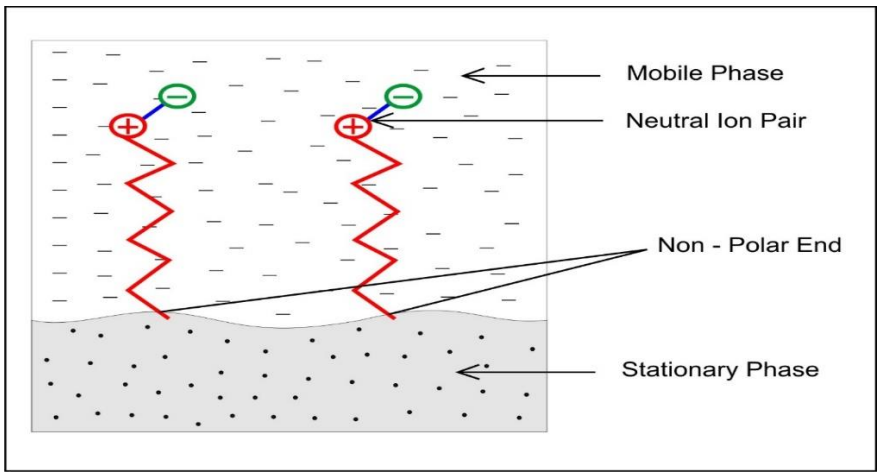


Fig.1.5 Reverse Phase Ion Pair Chromatography

5) Affinity chromatography:

Affinity chromatography is a method of separating a biomolecule from a mixture, based on a highly specific macromolecular binding interaction between the biomolecule and another substance. The specific type of binding interaction depends on the biomolecule of interest; antigen and antibody, enzyme and substrate, receptor and ligand,or protein and nucle ic acid binding interactions are frequently exploited for isolation of various biomolecules. Affinity chromatography is useful for its high selectivity and resolution of separation, compared to other chromatographic methods.

Affinity chromatography uses highly specific biochemical interactions for separation. The stationary phase contains specific groups of molecules which can absorb the sample if certain steric and charge related conditions are satisfied. This technique can be used to isolate proteins, enzymes as well as antibodies from complex mixtures.

Affinity chromatography has the advantage of specific binding interactions between the analyte of interest (normally dissolved in the mobile phase), and a binding partner or ligand (immobilized on the stationary phase). In a typical affinity chromatography experiment, the ligand is attached to a solid, insoluble matrix-usually a polymer such as agarose or polyacrylamide-chemically modified to introduce reactive functional groups with which the ligand can react, forming stable covalent bonds. The stationary phase is first loaded into a column to which the mobile phase is introduced.

Molecules that bind to the ligand will remain associated with the stationary phase. A wash buffer is then applied to remove non-target biomolecules by disrupting their weaker interactions with the stationary phase, while the biomolecules of interest will remain bound. Target biomolecules may then be removed by applying a so-called elution buffer, which disrupts interactions between the bound target biomolecules and the ligand. The target molecule is thus recovered in the eluting solution.

Affinity chromatography does not require the molecular weight, charge, hydrophobicity, or other physical properties of the analyte of interest to be known, although knowledge of its binding properties is useful in the design of a separation protocol.

Affinity chromatography can be used in a number of applications, including nucleic acid purification, protein purification from cell free extracts, and purification from blood.

By using affinity chromatography, one can separate proteins that bind to a certain fragment from proteins that do not bind that specific fragment. Because this technique of purification relies on the biological properties of the protein needed, it is a useful technique and proteins can be purified many folds in one step. Affinity chromatography offers high selectivity, resolution, and capacity in most protein purification schemes. It has the advantage of utilizing a protein's biological structure or function for purification. As a result, purifications that would otherwise be time consuming and complicated, can often be easily achieved with affinity chromatography.

Its major application includes:

- Separation of mixture of compounds.
- Removal of impurities or in purification process.
- In enzyme assays.
- Detection of substrates.
- Investigation of binding sites of enzymes.
- In in vitro antigen-antibody reactions.
- Detection of Single Nucleotide polymorphisms and mutations in nucleic acids

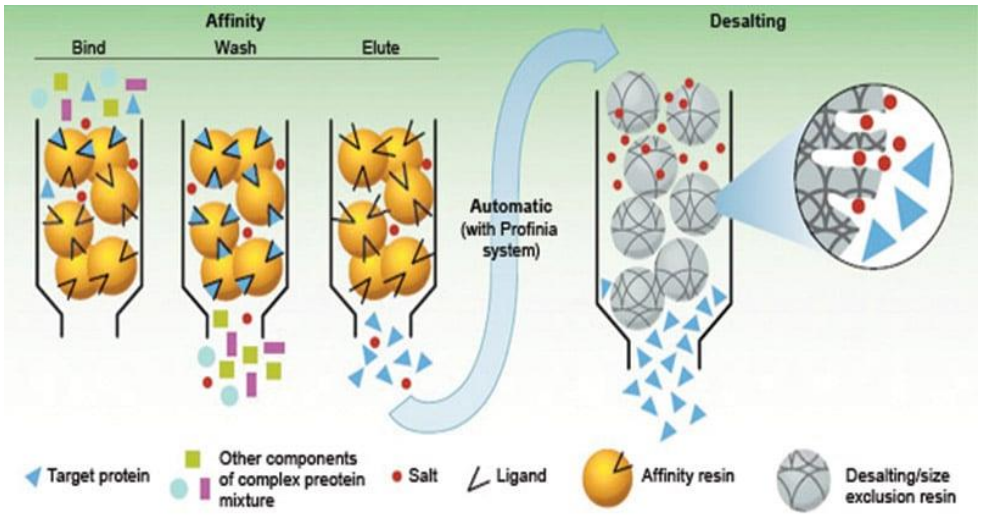


Fig.1.6 Affinity Chromatography

6) Size exclusion chromatography:

Size exclusion chromatography (SEC) separates molecules based on their size by filtration through a gel. The gel consists of spherical beads containing pores of a specific size distribution. Separation occurs when molecules of different sizes are included or excluded from the pores within the matrix. Small molecules diffuse into the pores and their flow through the column is retarded according to their size, while large molecules do not enter the pores and are eluted in the column's void volume. Consequently, molecules separate based on their size as they pass through the column and are eluted in order of decreasing molecular weight (MW).

Operating conditions and gel selection depend on the application and the desired resolution. Two common types of separations performed by SEC are fractionation and desalting (or buffer exchange).

Size exclusion chromatography separates molecules according to their molecular mass. Largest molecules are eluted first and the smallest molecules last. This method is generally used when a mixture contains compounds with a molecular mass difference of at least 10%.

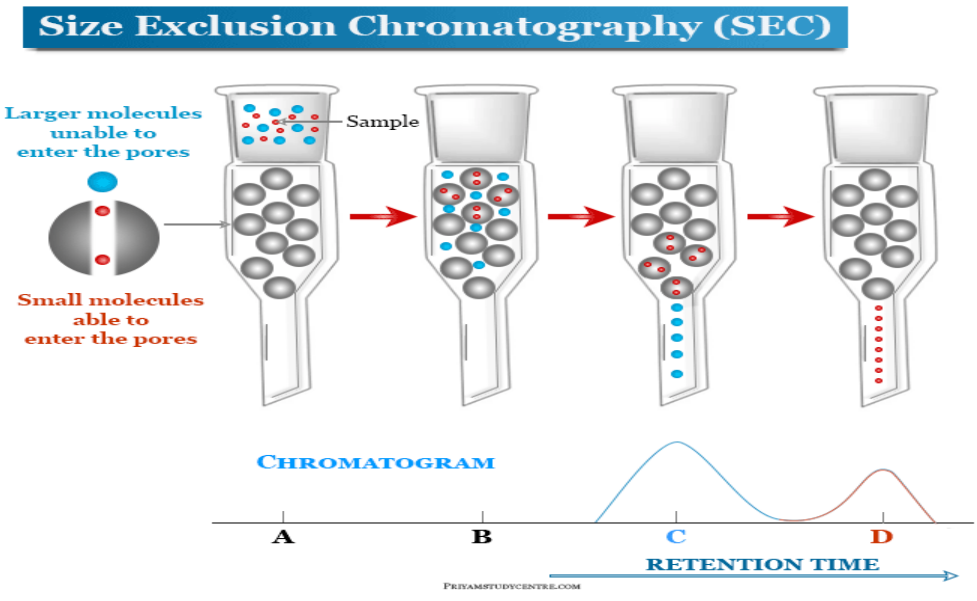


Fig.1.7 Size Exclusion Chromatography

This mode can be further subdivided into :-

i. Gel Permeation Chromatography
ii. Gel Filtration Chromatography

i.Gel Permeation Chromatography :

Gel permeation chromatography (**GPC**) is a type of size-exclusion chromatography (SEC), that separates high molecular weight or colloidal analytes on the basis of size or diameter, typically in organic solvents. The technique is often used for the analysis of polymers. As a technique, SEC was first developed in 1955 by Lathe and Ruthven.

The term gel permeation chromatography can be traced back to J.C. Moore of the Dow Chemical Company who investigated the technique in 1964. The proprietary column technology was licensed to Waters Corporation, who subsequently commercialized this technology in 1964. GPC systems and consumables are now also available from a number of manufacturers. It is often necessary to separate polymers, both to analyze them as well as to purify the desired product.

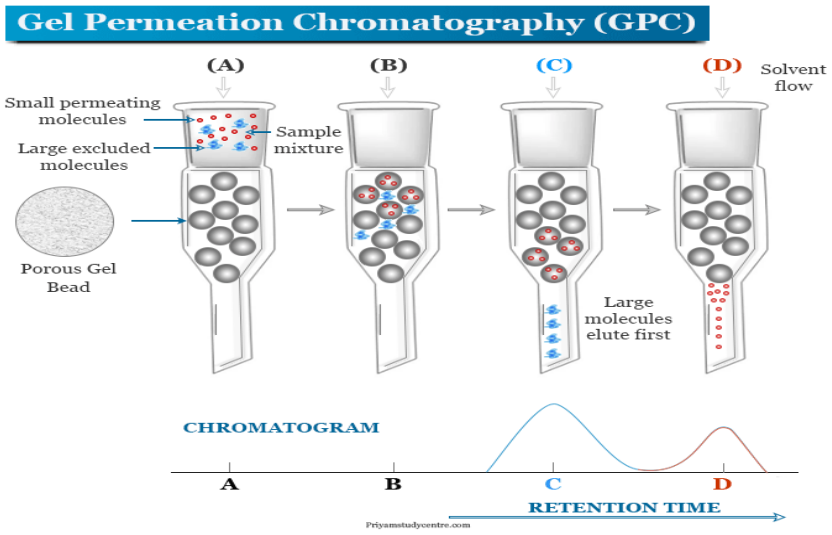


Fig.1.7.1 Gel Permeation Chromatography

ii.Gel Filtration Chromatography :

Gel filtration chromatography, a type of size exclusion chromatography, can be used to either fractionate molecules and complexes in a sample into fractions with a particular size range, to remove all molecules larger than a particular size from the sample, or a combination of both operations. Gel filtration chromatography can be used to separate compounds such as small molecules, proteins, protein complexes, polysaccharides, and nucleic acids when in aqueous solution. When an organic solvent is used as the mobile phase, the process is instead referred to as gel permeation chromatography.

Gel filtration chromatography can also be used for:

- Fractionation of molecules and complexes within a predetermined size range
- Size analysis and determination
- Removal of large proteins and complexes
- Buffer exchange

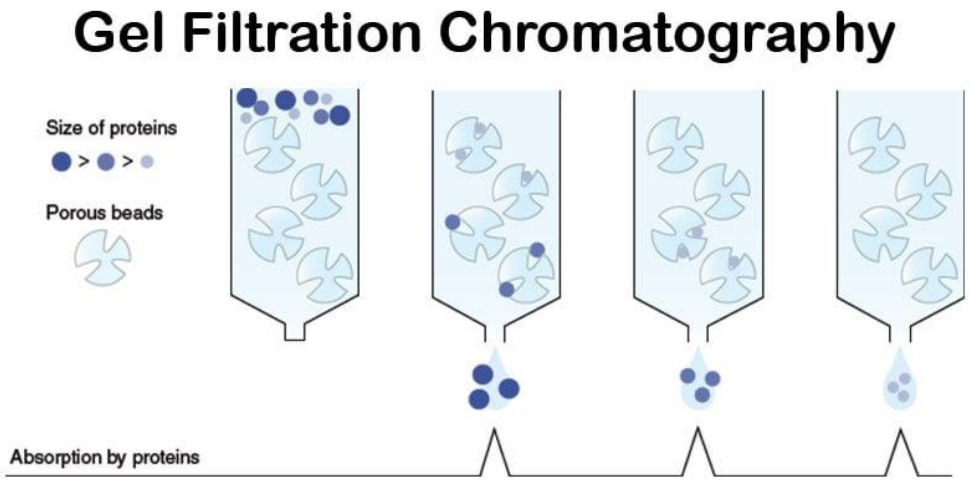


Fig.1.7.2 Gel Filtration Chromatography

Parameters that are affected by the changes in chromatographic conditions:

1. Resolution (Rs).

2. Capacity factor (k').

3. Selectivity (α),

4. Column efficiency (N).

5. Peak asymmetry factor (As).

1) Resolution (Rs):

Resolution is the parameter describing the separation power of the complete chromatographic system relative to the particular components of the mixture. The resolution (Rs), of two neighbouring peaks is defined as the ratio of the distance between two peak maxima. It is the difference between the retention times of two solutes divided by their average peak width. For baseline separation, the ideal value of Rs is 1.5. It is calculated by using the formula,

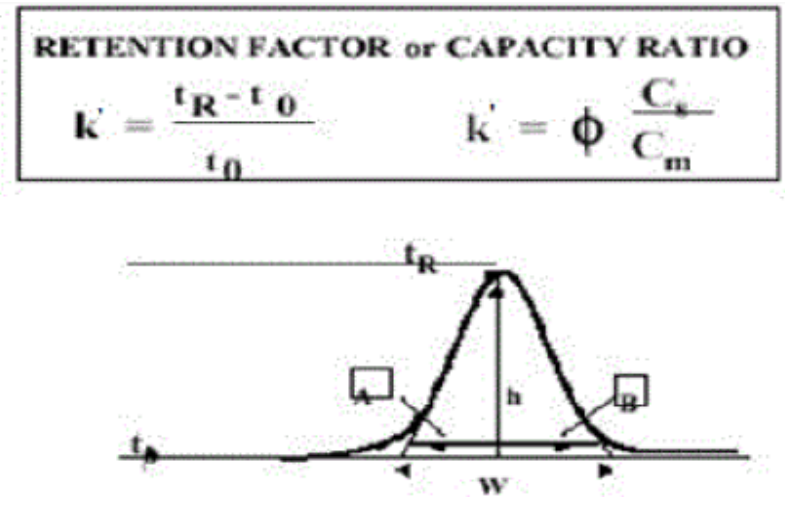


Fig.1.8. Resolution Between two peaks

Where,

$t_R(1)$ and $t_R(2)$ are the retention times of components 1 and 2 and

W_1 and W_2 are peak width of components 1 and 2.

Baseline resolution is achieved when R = 1.5

It is useful to relate the resolution to the number of plates in the column, the selectivity factor and the retention factors of the two solutes

$$R=\sqrt{N}/4(\alpha-1/\alpha)(1+ k'_B/ k'_B)$$

To obtain high resolution, the three terms must be maximized. An increase in N, the number of theoretical plates, by lengthening the column leads to an increase in retention time and increased band broadening which may not be desirable. Instead, to increase the number of plates, the height equivalent to a theoretical plate can be reduced by reducing the size of the stationary phase particles.

It is often found that by controlling the capacity factor (k'), separations can be greatly improved. This can be achieved by changing the temperature (in Gas Chromatography) or the composition of the mobile phase (in Liquid Chromatography).

2) Capacity factor (k'):

Capacity factor is the ratio of the reduced retention volume to the dead volume. Capacity factor (k'), is defined as the ratio of the number of molecules of solute in the stationary phase to the number of molecules of the same in the mobile phase. Capacity factor is a measure of how well the sample molecule is retained by a column during an isocratic separation. The ideal value of k' ranges from 2-10. Capacity factor can be determined by using the formula.

RETENTION FACTOR or CAPACITY RATIO

$$k' = \frac{t_R - t_0}{t_0} \qquad k' = \phi \frac{C_s}{C_m}$$

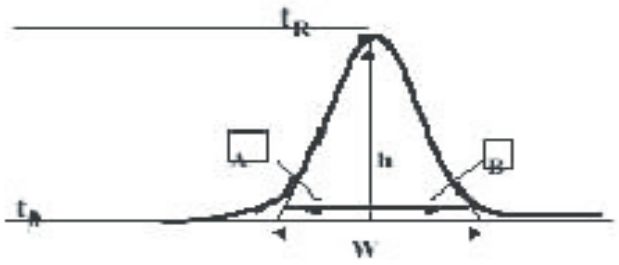


Fig.1.9 Capacity Factor

Where,

t_R = retention volume at the apex of the peak (solute)

t_0 = void volume of the system.

3) Selectivity factor (a):

It can also be manipulated to improve separations. When is close to unity, optimizing k' and increasing N is not sufficient to give good separation in a reasonable time. In these cases, k' is optimized first, and then it is increased by one of the following procedures:

1. Changing mobile phase composition.

2. Changing column temperature.

3. Changing composition of stationary phase.

4. Using special chemical effects (such as incorporating a species which complexes with one of the solutes into the stationary phase).

4) Column Efficiency (N):

Efficiency (N), of a column is measured by the number of theoretical plates per meter. It is a measure of band spreading of a peak. Similar the band spread, higher is the number of theoretical plates, indicating good column and system performance. Columns with N ranging from 5,000 to 1,00,000 plates/meter are ideal for a good system. Efficiency is calculated by using the formula,

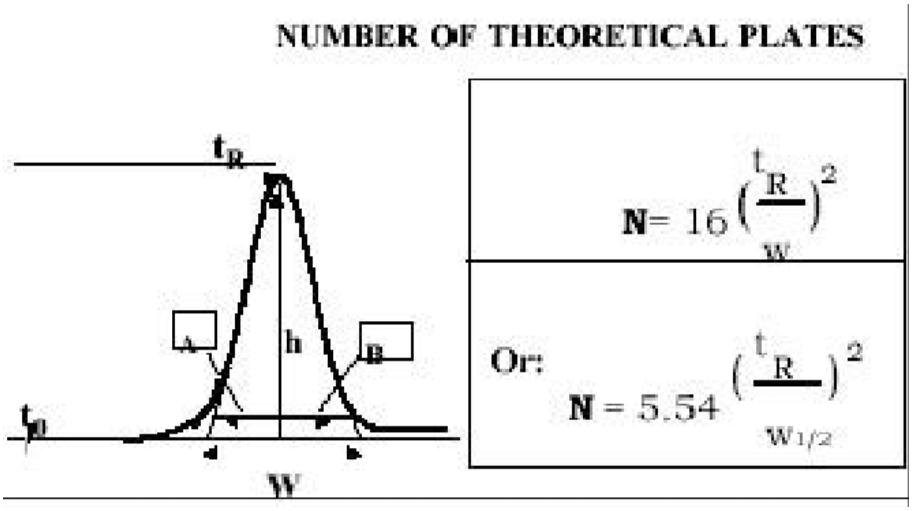


Fig.1.10 Number of Theoretical Plates

Where,

t_R is the retention time.

W is the peak width.

5) Peak asymmetry factor (Tf):

Peak asymmetry factor, (Tf) can be used as a criterion of column performance. The peak half width (b), of a peak at 10% of the peak height, divided by the corresponding front half width (a), gives the asymmetry factor.

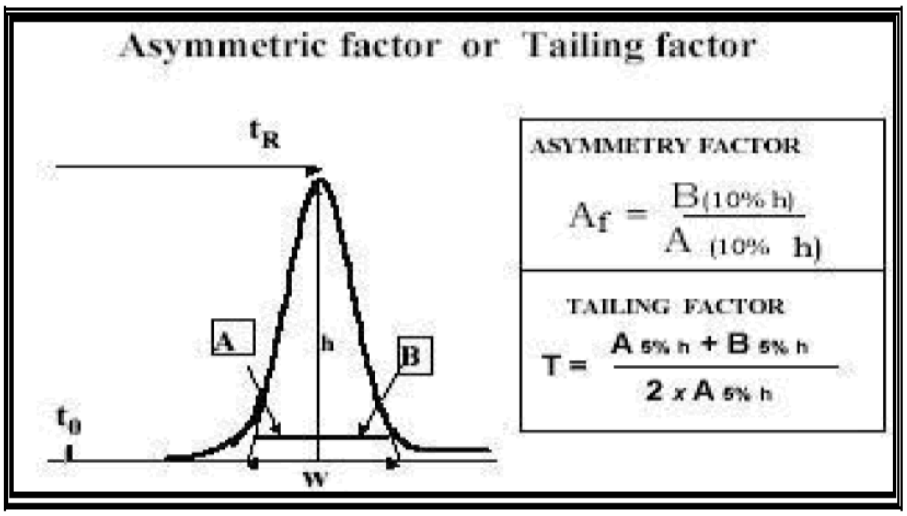


Fig.1.11 Assymetric Factor

For a well packed column, an asymmetry factor of 0.9 to 1.1 should be achievable.

1.3.2 Analytical Method Validation

Method validation is the process used to confirm that the analytical procedure employed for a specific test is suitable for its intended use. Results from method validation can be used to judge the quality, reliability and consistency of analytical results; it is an integral part of any good analytical practice.

Analytical methods need to be validated or revalidated. Before their introduction into routine use.

Whenever the conditions change for which the method has been validated (e.g., an instrument with different characteristics or samples with a different matrix).

Whenever the method is changed and the change is outside the original scope of the method. The USP has published specific guidelines for method validation for compound evaluation.

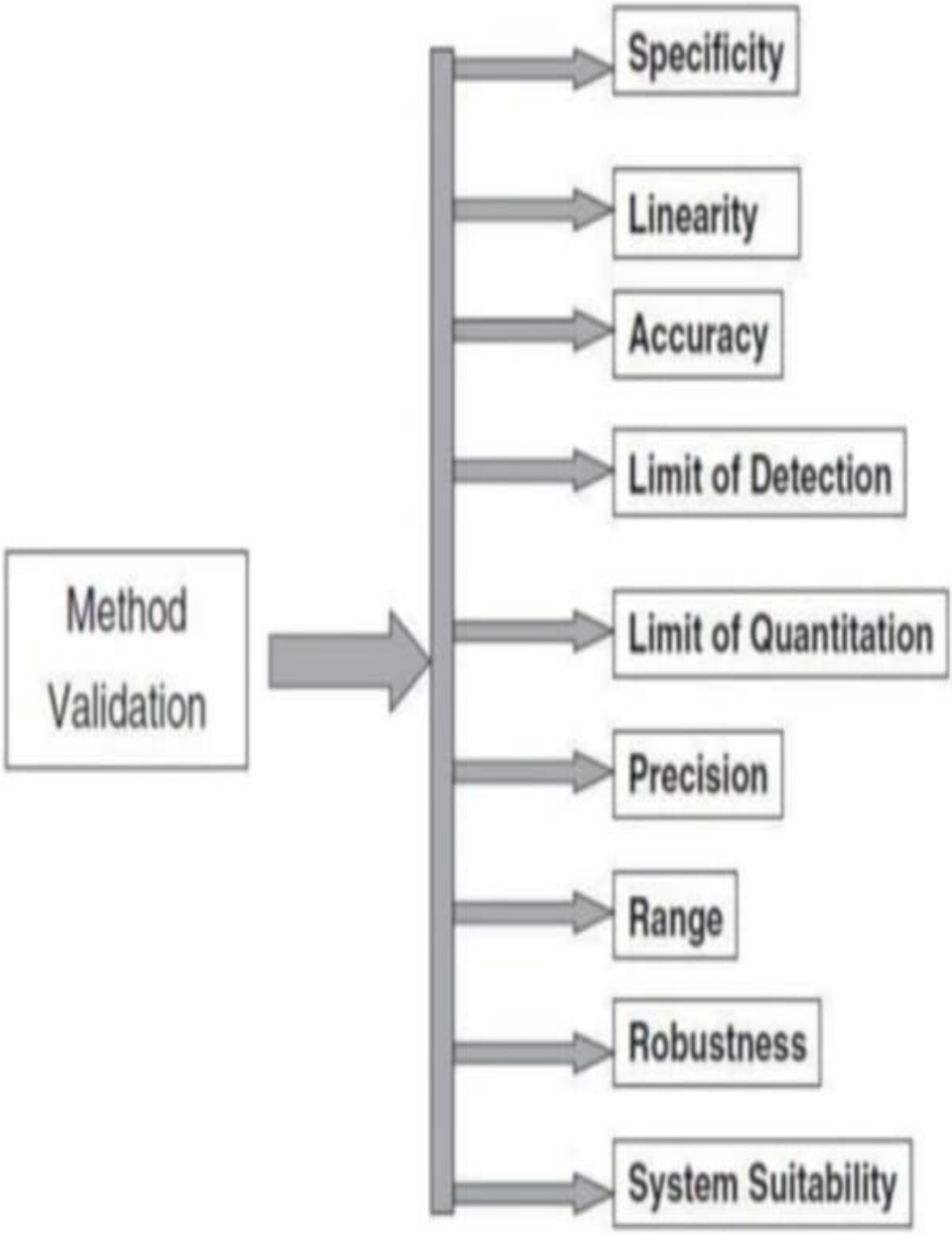


Fig.1.12 METHODS OF VALIDATION

ACCURACY

The accuracy of an analytical procedure expresses the closeness of agreement between the value which is accepted either as a conventional true value or an accepted reference value and the value found. This is sometimes termed trueness.

The accuracy of an analytical method should be established across its range. In the case of the assay of a drug in a formulated product, accuracy may be determined by application of the analytical method to synthetic mixtures of the drug product components to which known amount of analyte have been added within the range of the method. Minimum of test concentrations from 50% to 120% are normally used, for establishment of accuracy in assay of drug substance (or a finished product). Average recovery should be 98 to 102% of drug at each level.

PRECISION

The precision of an analytical procedure expresses the closeness of agreement (degree of scatter) between a series of measurements obtained from multiple sampling of the same homogeneous sample under the prescribed conditions. Precision may be considered at three levels: repeatability, intermediate precision and reproducibility.

Precision should be investigated using homogeneous, authentic samples. However, if it is not possible to obtain a homogeneous sample it may be investigated using artificially prepared samples or a sample solution.

The precision of an analytical procedure is usually expressed as the variance, standard deviation or coefficient of variation of a series of measurements. In the precision results of all samples should not have RSD >2%.

Reproducibility

Reproducibility expresses the precision between laboratories (collaborative studies, usually applied to standardization of methodology).

Determination of Reproducibility:

Reproducibility can be assessed by means of an inter-laboratory trial. Reproducibility should be considered in case of the standardization of an analytical procedure, for instance, for inclusion of procedures in pharmacopoeias.

Specificity

Specificity is the ability to assess unequivocally the analyte in the presence of components which may be expected to be present. Typically these might include impurities, degradants, matrix, etc.

Identification:

To ensure the identity of an analyte.

Purity Tests:

To ensure that all the analytical procedures performed allow an accurate statement of the content of impurities of an analyte, i.e. related substances test, heavy metals, residual solvents content, etc.

Assay:

To provide an exact result which allows an accurate statement on the content or potency of the analyte in a sample.

Determination of specificity:

ICH document state that when chromatographic procedure used, representative chromatograms should be used to demonstrate specificity and individual components should be appropriately detected. Peak purity tests may be useful to show that the analyte chromatographic peak is not attributable to more than one component (e.g., diode array, mass spectrometry).

Limit of Detection

The limit of detection of an individual analytical procedure is the lowest amount of analyte in a sample which can be detected but not necessarily quantitated as an exact value.

Determination of limit of detection:

For instrumental and non-instrumental methods detection limit is generally determined by the analysis of samples of known concentration of analyte and by establishing the minimum level at which the analyte can be reliability detected.

The limit of detection (LOD) may be expressed as:

LOD 3.3 /s

Where, the standard deviation of the response.

S the slope of the calibration curve.

The slope S may be estimated from the calibration curve of the analyte.

Limit of Quantitation

The limit of quantitation of an individual analytical procedure is the lowest amount of analyte in a sample which can be quantitatively determined with suitable precision and accuracy. The limit of quantitation is a parameter of quantitative assays for low levels of compounds in sample matrices, and is used particularly for the determination of impurities and/or degradation products.

Determination of limit of quantitation:

For instrumental and non-instrumental methods quantitation limit is generally determined by the analysis of samples of known concentration of analyte and by establishing the minimum level at which the analyte can be quantified with acceptable accuracy and precision.

The limit of quantitation (LOQ) may be expressed as:

LOQ =10 σ/s

Where,

Σ = the standard deviation of the response.

S = the slope of the calibration curve.

Linearity and Range

The linearity of an analytical procedure is its ability (within a given range) to obtain test results which are directly proportional to the concentration (amount) of analyte in the sample. The range of an analytical procedure is the interval between the upper and lower concentration (amounts) of analyte in the sample (including these concentrations) for which it has been demonstrated that the analytical procedure has a suitable level of precision, accuracy and linearity.

Determination of Linearity and Range:

For the determination of linearity, a minimum of 5 concentrations is recommended. Linearity can be determined by a series of sample whose concentrations span 80-120% of the expected concentration range. Linearity is evaluated by graphically.

Ruggedness

Degree of reproducibility of test results obtained by the same samples under a different condition such as, different analysts, different laboratories condition, different instrument etc. normally expressed as the lack of influence on test results of operational &environmental variables of the analytical method. Ruggedness is a measure of reproducibility of test results under the variation in the condition normally expected from laboratory to laboratory and from analyst to analyst.

Determination of Ruggedness:

By analysis of aliquots from homogenous lots in different laboratory, by different instrument and using operational and environmental condition that may differ but still with the specified parameters of the assay.

Degree of reproducibility of test results is then determined as a function of the assay variables.

i. Different operator in same laboratory
ii. Different equipment in same laboratory
iii. Different source of segment and solution
iv. Different source of column.

Robustness

The robustness of an analytical procedure is a measure of its capacity to remain unaffected by small, but deliberate variations in method parameters and provides an indication of its reliability during normal usage.

Determination of robustness:

The evaluation of robustness should be considered during the development phase and depends on the type of procedure under study. It should show the reliability of an analysis with respect to deliberate variations in method parameters.

Examples of typical variations are:

i. Stability of analytical solutions.
ii. Extraction time.

In the case of liquid chromatography, examples of typical variations are:

i. Influence of variations of pH in a mobile phase.
ii. Influence of variations in mobile phase composition.
iii. Different columns (different lots and/or suppliers).
iv. Temperature and flow rate.

Applications and Advantages

i. An ideal method for separation of various compounds in plant extracts which resemble in structure and thus demand specific and very sensitive method.

ii. A premier separation technique capable of multi component analysis of real life samples and complex mixtures.

iii. This method is used for ascertaining of various pharmaceuticals. The analysis of the various degradation products can be done and thus stability indicating HPLC systems and method has developed.

iv. Highly automated, using sophisticated auto-samplers and data systems for unattended analysis and report generation. Few techniques can match its versatility and precision of ±0.5% RSD.

V. A host of highly sensitive and specific detectors extend detection limits to nanogram, picogram, and even femtogram levels. As a preparative technique, it provides quantitative recovery of many labile components in milligram to kilogram quantities.

Introduction to Stability Degradation Method:-

Stability testing is termed as a complex process because of involvement of a variety of factors influencing the stability of a pharmaceutical product. These factors include stability of the active ingredient(s); interaction between active ingredients and excipients, manufacturing process followed, type of dosage form container/closure system used for packaging and light, heat and moisture conditions encountered during shipment, storage and handling. In addition, degradation reactions like oxidation, reduction, hydrolysis or racemization, which can play vital role in stability of a pharmaceutical product, also depend on such conditions like concentration of reactants, pH, radiation, catalysts etc., as well as the raw materials used and the length of time between manufacture and usage of the product.

A pharmaceutical product may undergo change in appearance, content uniformity, clarity (solution), moisture contents, particle size and shape, pH, package integrity thereby affecting its stability. Such physical changes may be because of impact, vibration, abrasion, and temperature fluctuations such as freezing, thawing or shearing etc. The chemical reactions

like solvolysis, oxidation, reduction, racemization etc. that occur in the pharmaceutical products may lead to the formation of degradation product, loss of potency of active pharmaceutical ingredient (API), loss of excipient activity like antimicrobial preservative action and antioxidants etc. (Carstensen et al., 2000). Stability of a pharmaceutical product can also be affected because of microbiological changes like growth of microorganisms in non sterile products and changes in preservative efficacy.

➤ Forced degradation study:

The major routes of degradation of any drug substance include hydrolysis, oxidation, heat and photolysis.

1. Hydrolytic degradation

2. Oxidative degradation

3. Thermal degradation

4. Photolytic degradation

1. Hydrolytic:

Hydrolytic study under acidic and basic condition involves catalyzation of ionisable functional groups present in the molecule. HCl and NaOH are employed for generating acidic and basic stress samples, respectively.

2. Oxidative Condition:

Many drug substances undergo autoxidation ie. oxidation under normal storage condition and involving ground state elemental oxygen.

Therefore it is an important degradation pathway of many drugs. Auto- oxidation is a free radical reaction that requires free radical initiator to begin the chain reaction. Hydrogen peroxide, metal ions, or trace level of impurities in a drug substance act as initiators for auto-oxidation.

The mechanism of oxidative degradation of drug substance involves an electron transfer mechanism to form reactive anions and cations. Amines, sulphides and phenols are susceptible to electron transfer oxidation to give N- oxides, hydroxylamine, sulphones and sulphoxide.

Hydrogen peroxide is very common oxidant to produce oxidative degradation products which may arise as minor impurities during long term stability studies.

3.Thermal Condition:

In general, rate of a reaction increase with increase in temperature. Hence, the drugs are susceptible to degradation at higher temperature.

Many APIs are sensitive to heat or tropical temperatures. For example, vitamins, peptides, etc. Thermal degradation involves different reactions like pyrolysis, hydrolysis, decarboxylation, isomerization, rearrangement and polymerization.

4. Photolytic Condition:

The rate of photodegradation depends upon the intensity of incident light and quantity of light absorbed by the drug molecule. The photolytic degradation can occur through non-oxidative or oxidative photolytic reaction.

Photolytic degradation is carried out by exposing the drug substance or drug product to a combination of visible and UV light.

The non-oxidative photolytic reaction include isomerization, dimerization, cyclization, rearrangements & decarboxylation etc. and while oxidative photolytic reaction occur through either singlet oxygen ($1O_2$) or triplet oxygen ($3O_2$) mechanism.

REVIEW OF LITERATURE

2. REVIEW OF LITERATURE

NEOSTIGMINE METHYLSULFATE

2.1 OFFICIAL METHOD FOR NEOSTIGMINE METHYLSULFATE

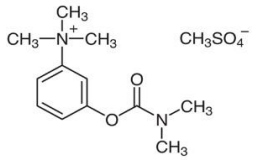
Sr. No	Official in	Method	Brief Introduction	Ref No
1	Neostigmine Methylsulfate	**UV Spectroscopy**	**Wavelength:-** 420nm **Concentration:-** 400µg/ml	11

2.2 REPORTED METHOD FOR NEOSTIGMINE METHYLSULFATE

Sr. No	Official in	Method	Brief Introduction	Ref No
1	Neostigmine Methylsulphate	**RP_HPLC**	**Mobile phase:-** 0.05M Phosphate buffer Acetonitrile (87:13) **Stationary phase:-** C_8 (250 X 4.6 , 5µ) **Flow rate:-** 1ml/min **Wavelength:-** 400µg/ml	12

2	Neostigmine Methylsulphate	**RP_HPLC**	**Mobile phase:-** 0.05M Phosphate buffer , pH 6,4: Methanol: Acetonitrile(75:10:15) **Stationary phase:-** C_{18} (250 X 4.6 , 5µ) **Flow rate:-** 1ml/min **Wavelength:-** 220nm	13
3	Glycopyrrolate And Neostigmine	**RP_HPLC**	**Mobile phase:-** Water : Acetonitrile(70:30) **Stationary phase:-** C_{18} (250 X 4.6 , 5µ) **Flow rate:-** 1ml/min **Wavelength:-** 255nm	14

2.3 DRUG PROFILE

Introduction	
Name	**Neostigmine methylsulfate**
Official in	USP30-NF25
Description	Neostigmine is a cholinesterase inhibitor used in the treatment of myasthenia gravis and to reverse the effects of muscle relaxants such as gallamine and tubocurarine. Neostigmine. unlike physostigmine, does not cross the blood-brain barrier
Structure	
Chemical Formula	$C_{13}H_{22}N_2O_6S$
Mol. Weight	334.40
IU PAC Name	3-[(dimethylcarbamoyl)oxy]-N.N.N-trimethylanilinium methylsulfate
Categories	Cholinergic Agents
Solubility	Soluble in water and organic solvents
Pharmacology	
Classes	Benzenoids
Mechanism of Action	Neostigmine is a parasympathomimetic, specifically, a reversible cholinesterase inhibitor. The drug inhibits acetylcholinesterase which is responsible for the degradation

	of acetylcholine. So, with acetylcholinesterase inhibited, more acetylcholine is present By interfering with the breakdown of acetylcholine, neostigmine indirectly stimulates both nicotinic and muscarinic receptors which are involved in muscle contraction.. It does not cross the blood-brain barrier.
Properties	
State	Solid
CAS No	51-60-5
Melting point	142-146º C

2.4 DOSAGE FORM :

Brand name	Formulation	Contents	Manufacturer
Neostigmine methylsulfate injection	Injection	250mg / 100ml	Hamlen



AIM

AND

OBJECTIVE

3.AIM AND OBJECTIVE

❖ AIM

- To Develop stability indicating RP-HPLC Method for the Estimation of Neostigmine Methyl sulphate in its Pharmaceutical Dosage form.

❖ OBJECTIVE

- To develop HPLC method for estimation of Neostigmine Methyl sulphate in its Pharmaceutical Dosage form.
- To perform validation on the developed method.
- Applying the newly developed, validated analytical method for the estimation of Neostigmine Methyl sulphate in its Pharmaceutical Dosage form.

MATERIALS AND METHODS

4. MATERIALS AND METHODS

Table 4.1 Standards and Reagents

Standard	Source
Neostigmine Methylsulfate	Zydus

Table 4.2 Standards Used

Sample	Source
Neostigmine Methylsulfate Injection	Hamlen

Table 4.3 Samples Used

Chemical / Reagent	Grade	Manufacturer
Methanol	HPLC Grade	Merck specialities pvt, Ltd., Mumbai
Acetonitrile	HPLC Grade	Merck specialities pvt, Ltd., Mumbai
Water	HPLC Grade	Merck specialities pvt, Ltd., Mumbai

Table 4.4 Apparatus used in experiment

Components	Volume	Type
Volumetric Flasks	10ml, 25ml, 50ml, 100ml	Borosilicate Glass Type 1
Pipettes	1ml, 2ml, 5ml, 10ml	Borosilicate Glass Type 1
Measuring Cylinder	100ml	Borosilicate Glass Type 1
Beaker	100ml, 250ml, 500ml	Borosilicate Glass Type 1
Whatman Filter	-------	Filter Paper No.41

Table 4.5 Instrumentation

Component	Brand / Model / Software	Manufacturer / Supplier
HPLC	LC-20AT / Spinchrom	Shimadzu
HPLC Column	C18 (25cm x 0.46cm) Hypersil BDS	Cromasil
Detector	SPD-20A	Shimadzu
Ultrasonic Water Bath	----	Leelasonic

pH meter	pHcal	Analab
Analytical Balance	AX-200	Shimadzu
UV Visible spectrophotometer	UV spectrophotometer 119	Systronic
Melting point Apparatus	Thermocal	Analab

Table 4.6 Identification of drugs

Determination of Solubility

Drug	Solubility
Neostigmine Methylsulfate	Freely soluble in water and methanol

Table 4.7 Determination of Melting point

Drug	Standard Melting Point	Observed Melting Point
Neostigmine Methylsulfate	142-146°c	145-147°c

4.8 Selection of wavelength

The sensitivity of HPLC method that uses UV detection depends upon proper selection of detection wavelength. An ideal wavelength is the one that gives good response for the drugs that are to be detected. In the present study drug solution of Neostigmine Methylsulfate (20µg/ml) was prepared in Methanol. This drug solution was than scanned in UV region of 200-400 nm and overlay spectrums were recorded.

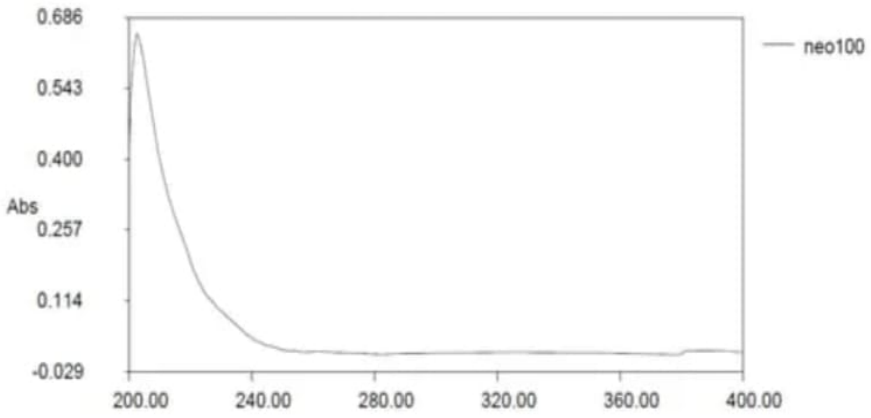


Fig.4.8 : UV Spectra of Neostigmine Methylsulfate (100µg/ml) in Methanol

Neostigmine Methylsulfate solution: 100 mg-100ml with methanol. Further Iml to a 10ml and make up with methanol (100µg/ml in methanol)

This solution was scanned between 200-400 nm. Wavelength was selected from the spectra of above solution.

4.9 Selection of Mobile Phase

Trail contains various mobile phase which are considered of Methanol, Water and Acetonitrile in different proportions and different volumes at different flow rate were tried. On the basis of various trails below given mobile phase, at 1.0 mL/min flow rate, proved to be better than the other mixture in terms of peak shape, theoretical plate and asymmetry.

Mobile phase A: 0.1% v/v perchloric acid

Mobile phase B: Acetonitrile: Water (95:5)

Time	Mob-A	Mob-B
0	85	15
8	85	15
10	45	55
11	70	30
12	85	15
15	85	15

Table 4.8. Analysis of marketed formulation by developed method

Sample Stock Solution (Neostigmine Methylsulfate 1000 µg/mL):

Take Injection equivalent to 100mg of Neostigmine Methylsulfate and transferred to a 100 ml volumetric flask. Add 60 ml Mobile phase. Shake for 15 min and make up volume with Mobile phase. The solution was filtered through Whatman filter paper no. 42.

4.10 Working Sample Preparation (Neostigmine Methylsulfate 100 µg/mL):

Take 1 ml. from standard stock solution and transferred to 10 ml volumetric flask and made up volume up to the mark with the mobile phase.

Inject above Solution 20 µl for Assay Analysis.

Injection	Neostigmine Methylsulfate Injection
Label claim	Neostigmine Methylsulfate (1% w/v)
Assay (% of label claim*)	98.252

Table 4.10 Working Sample Preparation

RESULT AND DISCUSSIONS

5. RESULT AND DISCUSSION

5.1 Trails Of Summaries

SR.no	Mobile phase	Remark
1	Water Methanol(50:50)	No peak observed
2	Water Acetonitrile(50:50)	Still no peak observed
3	Water PH3.0:Acetonitrile(50:50)	Peak observed at solvent peak time
4	Water pH3.0:Acetonitile(60:40)	Still peak observed at solvent peak tome
5	Water pH 3.0:acetonitrile(85:15)	Retention time increased
6	Gradient programme	Gradient programme set to resolve placebo from the neostigmine peak

5.2. List of Mobile Phase trails

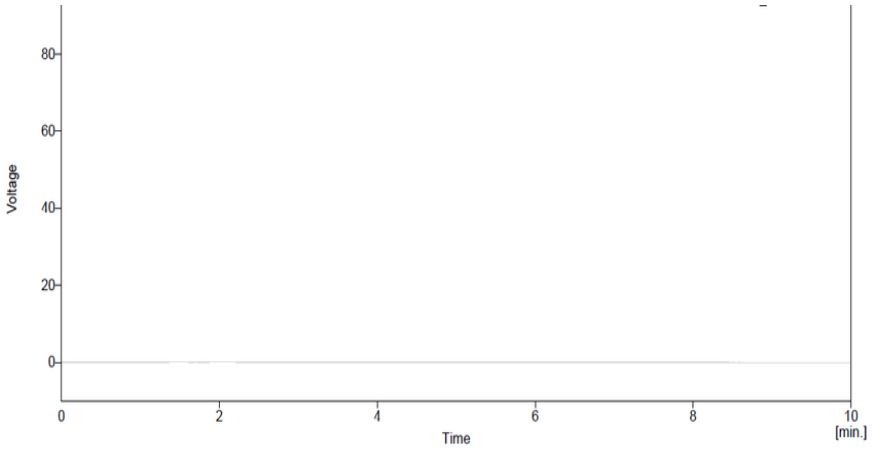


Fig.5.2.1 HPLC Chromatogram of Neostigmine Methylsulfate100µg / ml in Water : Methanol (50 : 50)

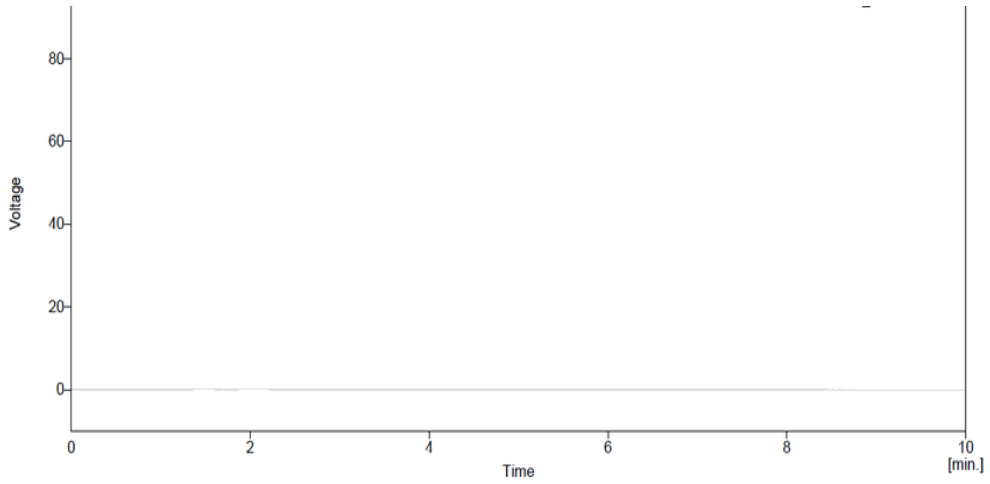


Fig.5.2.2. HPLC Chromatogram of neostigmine methylsulfate 100µg/ml in water : Acetonitrile (50 : 50)

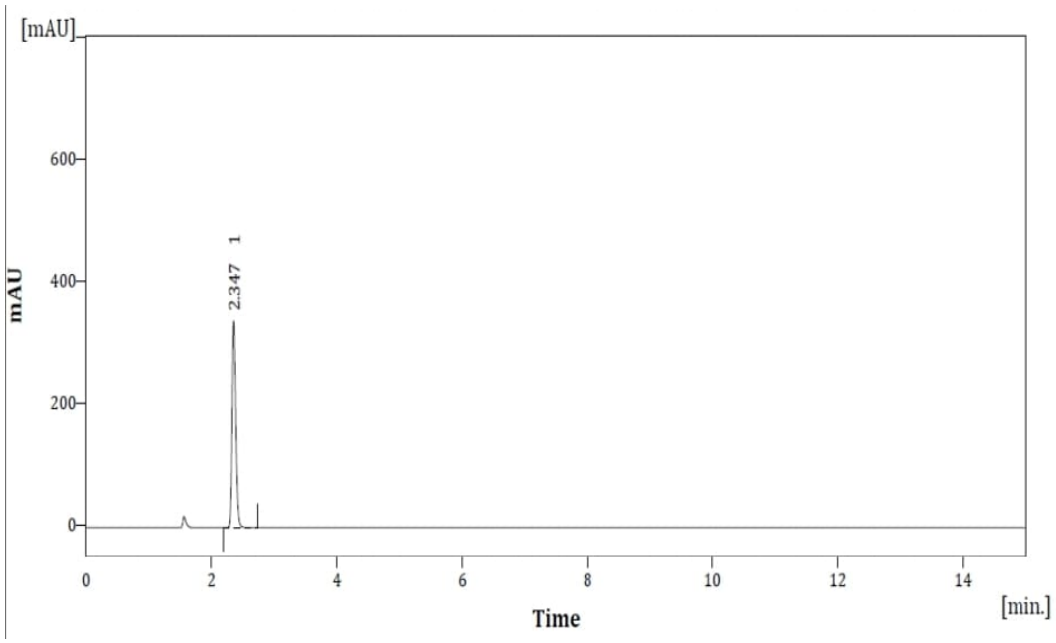


Fig.5.2.3. HPLC Chromatogram of neostigmine methylsulfate 100µg/ml in water, pH 3.0:Acetonitrile (50 : 50)

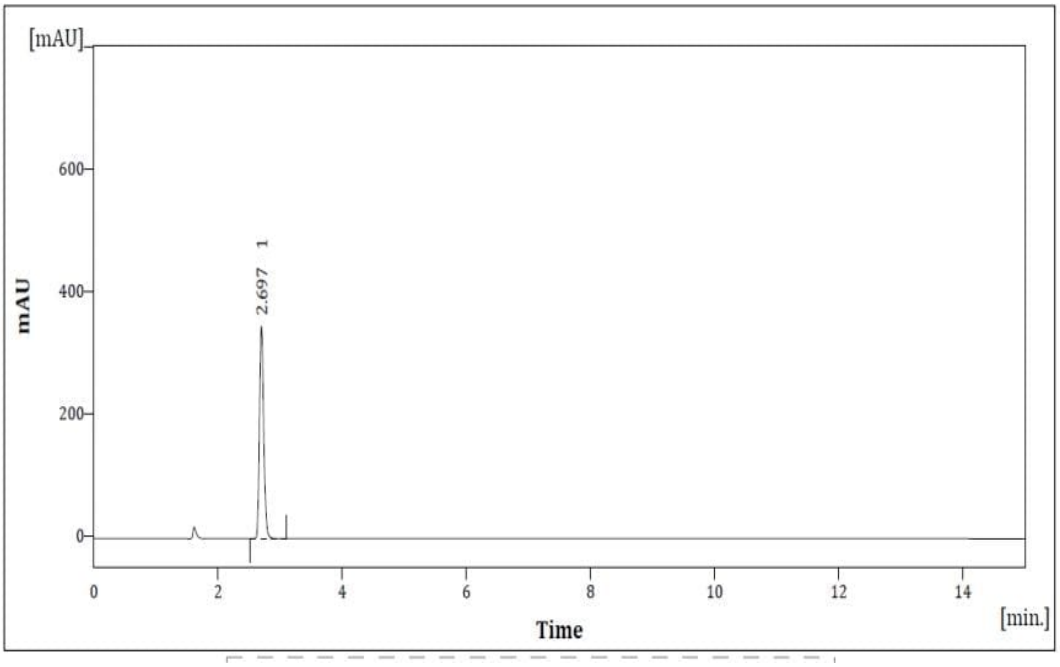


Fig.5.2.4. HPLC Chromatogram of neostigmine methylsulfate 100µg/ml in water, pH 3.0:Acetonitrile (60 : 40)

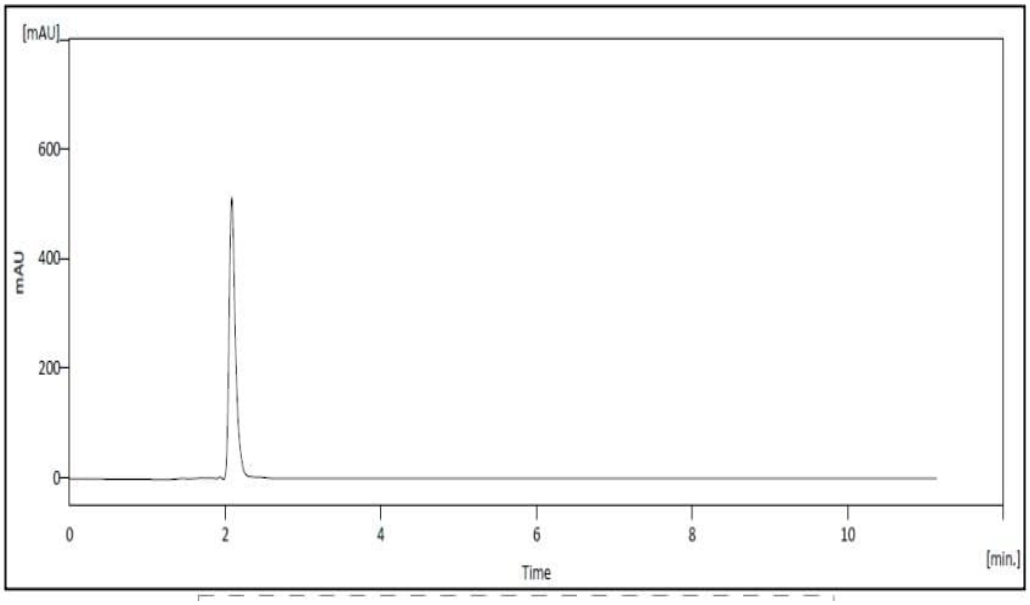


Fig.5.2.5. HPLC Chromatogram of neostigmine methylsulfate 100µg/ml in water, pH 3.0:Acetonitrile (85 : 15)

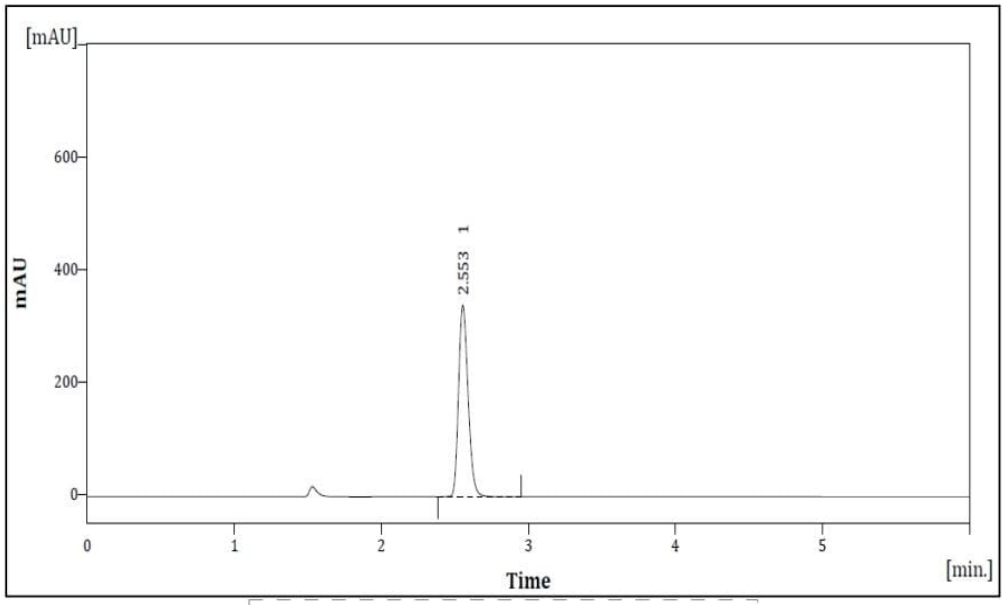


Fig.5.2.6. HPLC Chromatogram of neostigmine methylsulfate 100µg/ml in gradient programme

Table 5.3 System suitability

Parameters	**Neostigmine Methylsulfate**
Retention Time	3.123
Theoretical Plates	7782
Asymmetry	1.368

Table 5.4 Final chromatographic condition

Parameters	Chromatographic Condition
Mode of elution	Gradient
Mobile Phase	Mobile phase A: 0.1% v/v perchloric acid Mobile phase B: Acetonitrile Water (95.5)
Column	C18 (25cm x 0.46 cm) Hypersil BDS
Flow rate	1ml/min
Runtime	20min
Injection volume	20µL
Detection wavelength	210nm

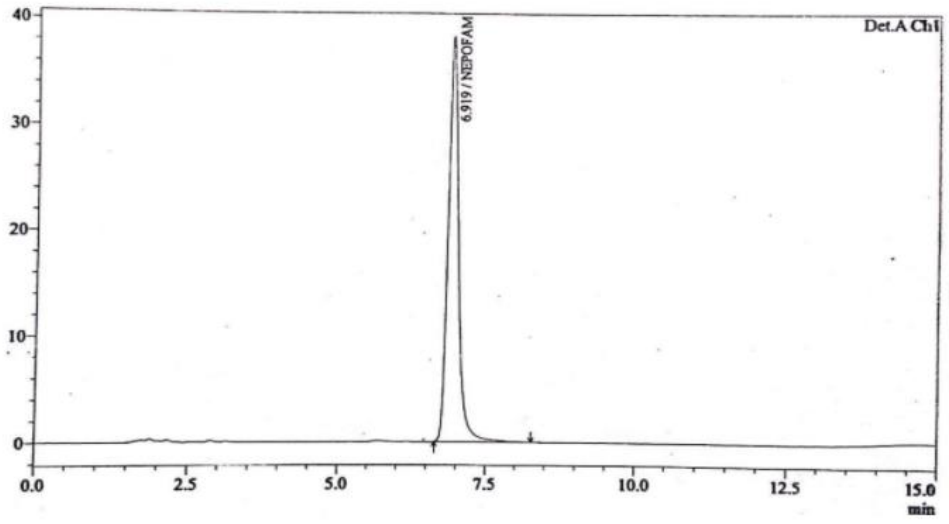
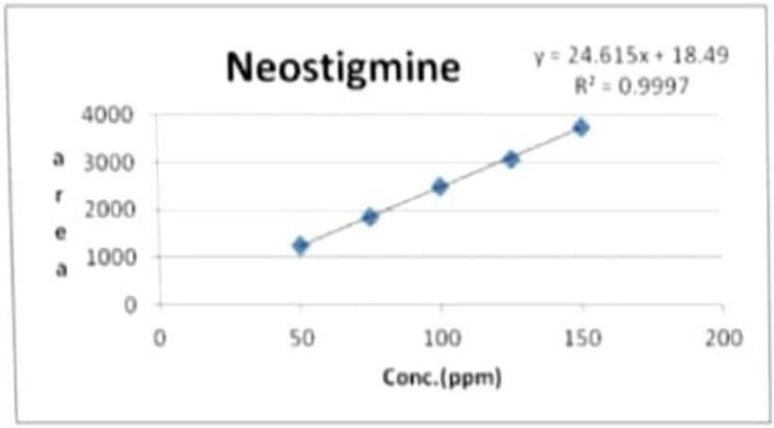


Fig.5.5 Overlay chromatogram of different concentration of Neostigmine Methylsulfate



Interday precision

Standard solution containing (50, 100, 150µg/ml) of Neostigmine Methylsulfate was analysed three times on the different day and % R.S.D was calculated

	Neostigmine Methylsulfate		
SR.no	**Conc.(µg/ml)**	**Area Mean ± S.D.(n=3)**	**% R.S.D**
1	50	2488.448 ± 13.649	0.584
2	100	2466.218 ± 21.080	0.855
3	150	3716.611 ± 21.514	0.579

Table 5.6 Interday precision data for estimation of Neostigmine Methylsulfate

5.7 LOD and LOQ:

Calibration curve was repeated for five times and the standard deviation (SD) of the intercepts was calculated. Then LOD and LOQ were calculated as follows:

LOD-3.3 SD/slope of calibration curve

LOQ10 SD/slope of calibration curve

Where, SD = Standard deviation of intercepts

Neostigmine Methylsulfate
LOD = 3.3 x (SD / Slope) = 3.3 x (20.13/24.61) = 2.70 µg / ml

Limit of detection data for Neostigmine Methylsulfate

Neostigmine Methylsulfate
LOQ = 10 x (SD / Slope) = 10 x (20.13/24.61) = 8.10 µg / ml

Limit of quantitation data for Neostigmine Methylsulfate

DISCUSSION

We can Decided use of chromatogram drugs on different Concentration 6 trials of Concentration gradient and finally we got the Neo stigmine methyle sulfonate drug Peak and Calibration Curve Calculation. Mobile Phase was selected based on the review of literature. Various mobile phases were tried. Trial contains various mobile phases which consisted of Methanol, Water and Acetonitrile in different proportions at flow rate 1 ml/min were tried.

The Chromatograms of Neostigmine Methylsulfate standard and Neostigmine Methylsulfate sample show no interference with the Chromatogram of Blank, so the Developed method is Specific

The assay results were comparable to labelled value of drug in dosage form. These results indicate that the developed method is precise, simple and rapid. It can be used in the routine quality control of dosage form in industries. Iml/min flow rate, proved to be better than the other in terms of resolution, peak shape and shorter retention time.

- The linearity for Neostigmine Methylsulfate was assessed by analysis of standard solution in range of 50-150µg/ml.
- 5,7.5,10,12.5,15 ml solutions were pipette out from the Stock solution of Neostigmine Methylsulfate (1000 µg/ml) and transfer to 100 ml volumetric flask and make up with mobile phase to obtain 50,75,100,125 and 150 µg/ml for Neostigmine Methylsulfate
- In term of slope, intercept and correlation co-efficient value, The graph of peak area obtained verses respective concentration was plotted.
- Correlation co-efficient for calibration curve Neostigmine Methylsulfate was found to be 0.9997.
- The regression line equation for Neostigmine Methylsulfate is y24.615x + 18.49

CONCLUSION

6. CONCLUSION

- A simple, specific, linear and precise RP-HPLC method has been developed and validated as per ICH guideline for Estimation of Neostigmine Methylsulfate in its dosage form.
- Validation parameters like Specificity, Linearity and Precision were tested.
- Observation of all these parameters leads to the point that developed RP-HPLC method is simple, specific, linear and precise.
- It can be successfully adopted for routine quality control analysis of Neostigmine Methylsulfate in its dosage form without any interference from common excipients and impurity.
- This method can now transfer to utilize for routine laboratory analysis and assay of Neostigmine Methylsulfate in its dosage form.

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